

ENV 6106

Air Pollution Modeling

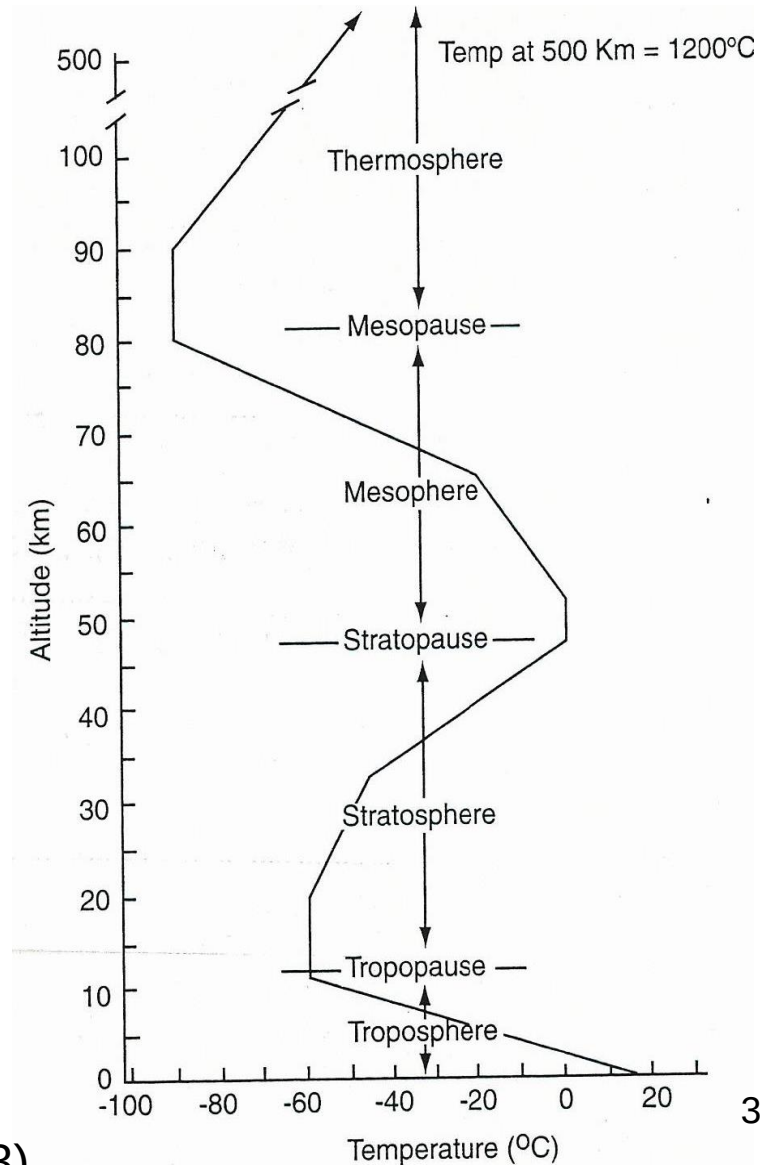
Spring 2023

Topics

- Basic meteorology related to air quality
- Expectation
 - Refresh knowledge on atmosphere motion at different scales
 - Understand the structure (density, pressure, temperature) of atmosphere, lapse rate (dry, wet and environmental), inversion, stability class
 - Be able to convert temperature and pressure among different units
- Readings: Cooper & Alley Chapter 19.1-19.4

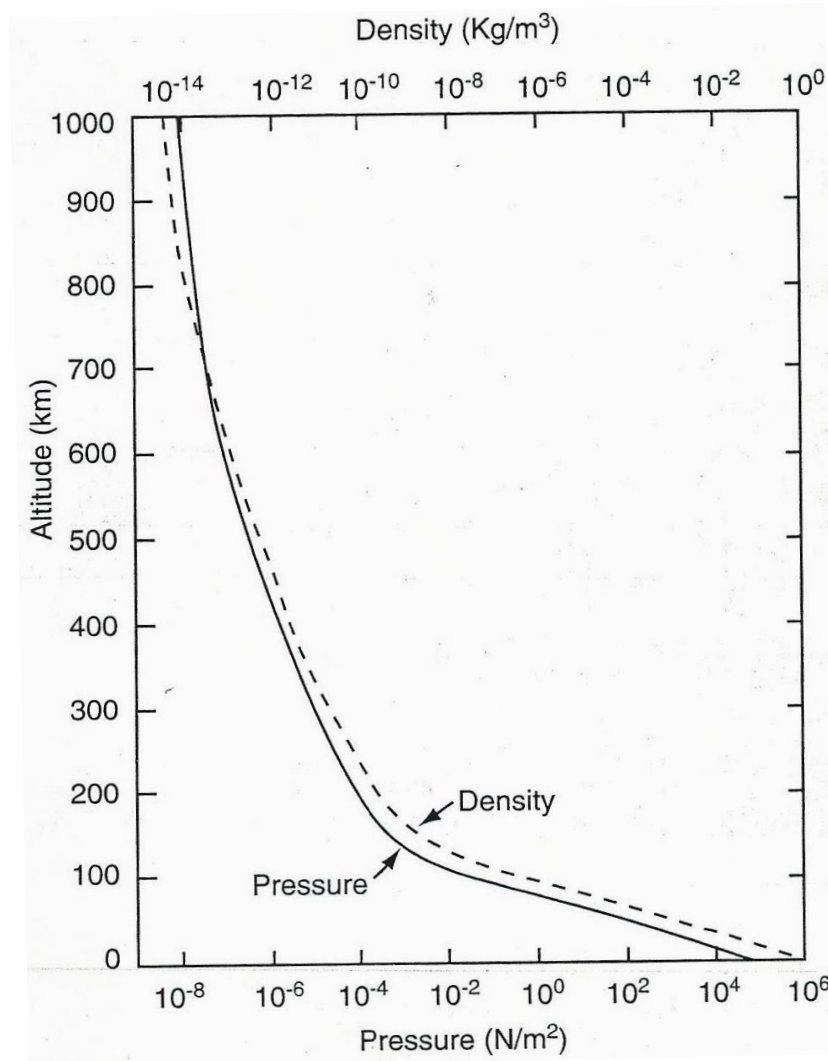
Atmospheric Temperature Structure

- Troposphere
 - Height ↑ Temp ↓
 - Rapid vertical mixing
- Stratosphere
 - Height ↑ Temp ↑
 - Inversion
 - Stable



(Godish, 2003)

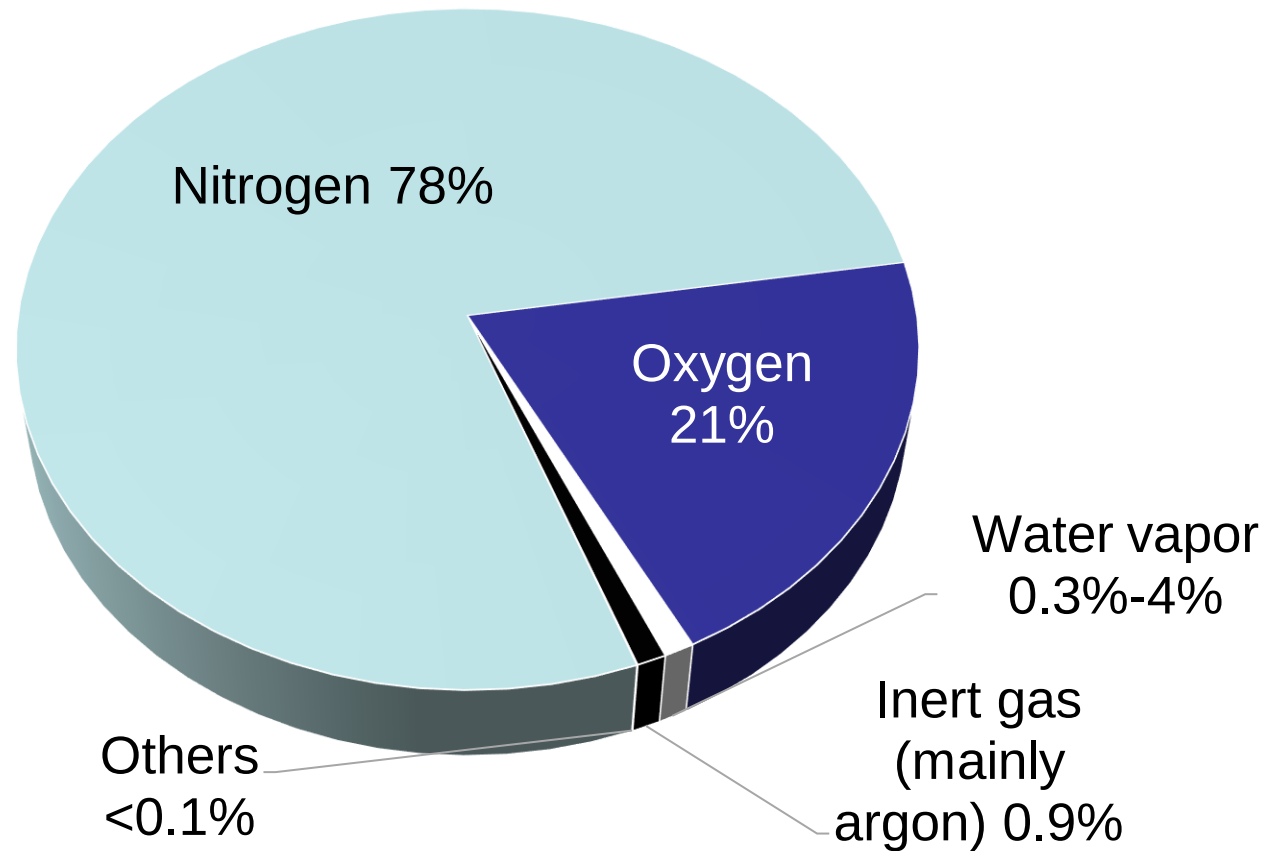
Atmospheric Density and Pressure



Conversion of temperature & pressure

- Temperature: Kelvin (K), Celsius (°C), Fahrenheit (°F)
 - $K = 273.15 + ^\circ\text{C}$
 - $^\circ\text{F} = 1.8 \times ^\circ\text{C} + 32$
 - $^\circ\text{C} = (^\circ\text{F} - 32)/1.8$
- Pressure
 - $1 \text{ atm} = 1013.25 \text{ mbars} = 1.01325 \times 10^5 \text{ Pa}$
 $= 14.69 \text{ lb/in}^2 \text{ (psi)} = 760 \text{ mmHg}$
 $= 10332 \text{ mmH}_2\text{O}$

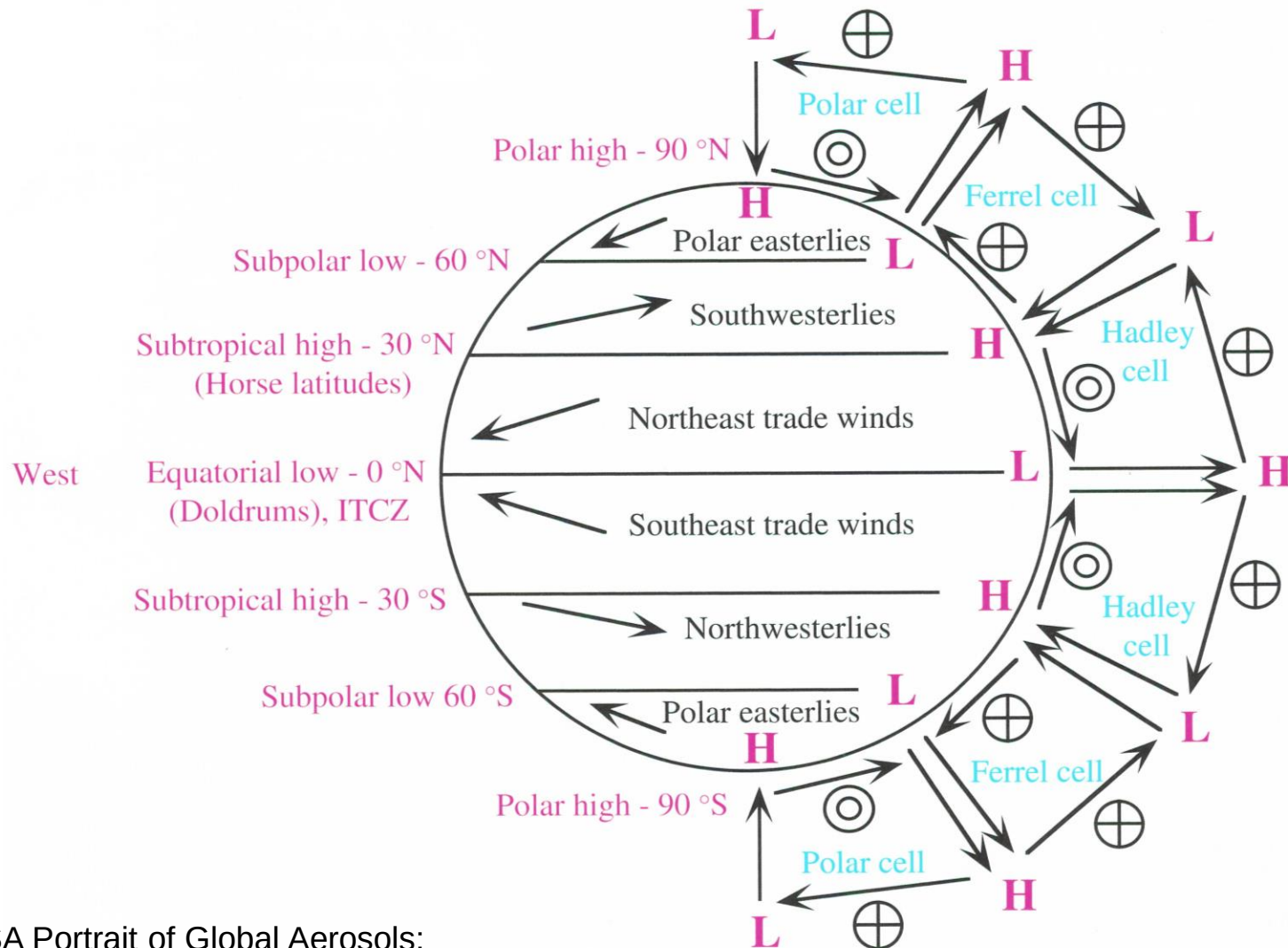
Composition of the Troposphere



Pollution transport

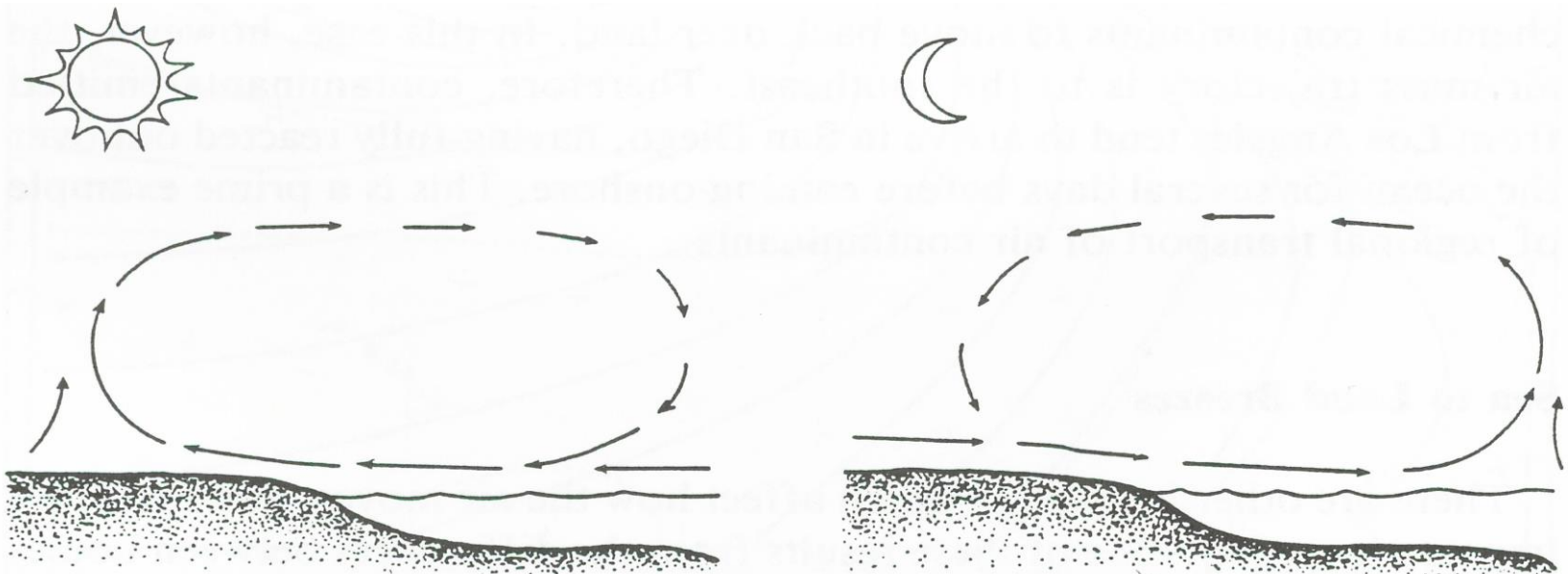
- After emitted/formed, pollutants need to be transported to get in contact with “receptor”
 - Human, certain part of eco-system etc..
- Pollution transport depends on meteorology
- Pollutants can be removed or formed during the process
- The distance of transport differ drastically depends on pollutant characteristics

Large Scale Motion: General Circulation pattern



Smaller scale motion

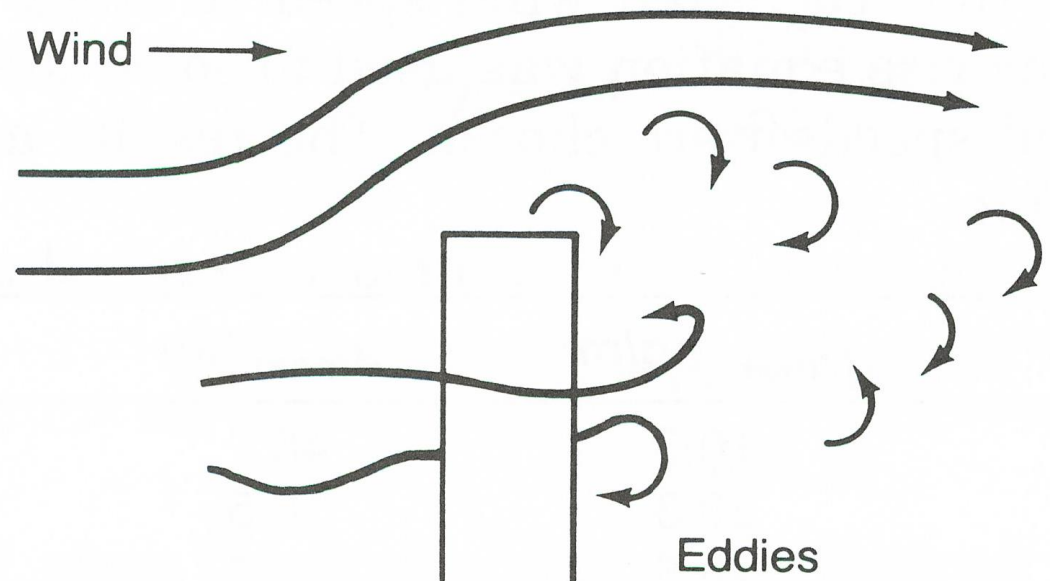
- Impact by variable heating and topography
 - valley and mountain breezes
 - sea and land breezes
 - urban heat island flow



Example of land sea breeze (Griffin, 1994)

Even smaller scale

- Turbulence
 - Random / irregular fluctuations on top of mean flow
 - Circular eddies at smaller scale than mean flow
- Important to
 - mixing
 - dilution
- Pollution hot spot



Understanding Stability:

Vertical air motion

- Buoyancy
 - Air hotter than surrounding is less dense (and rises)
- Rising and falling air leads to
 - vertical overturning and mixing
 - pollutant dispersal/dilution due to turbulence
- Atmosphere can enhance or suppress this
 - Stable – atmosphere resists vertical motion
 - Unstable – atmosphere enhances vertical motion
 - Neutral – atmosphere does neither

Understanding Stability:

Vertical air motion

- Say we have a air parcel rise from the ground....
 - Stable → atmosphere hamper the movement
 - Unstable → atmosphere facilitate the movement
 - Neutral → atmosphere does neither
- The environmental temperature structure is important

Lapse rate

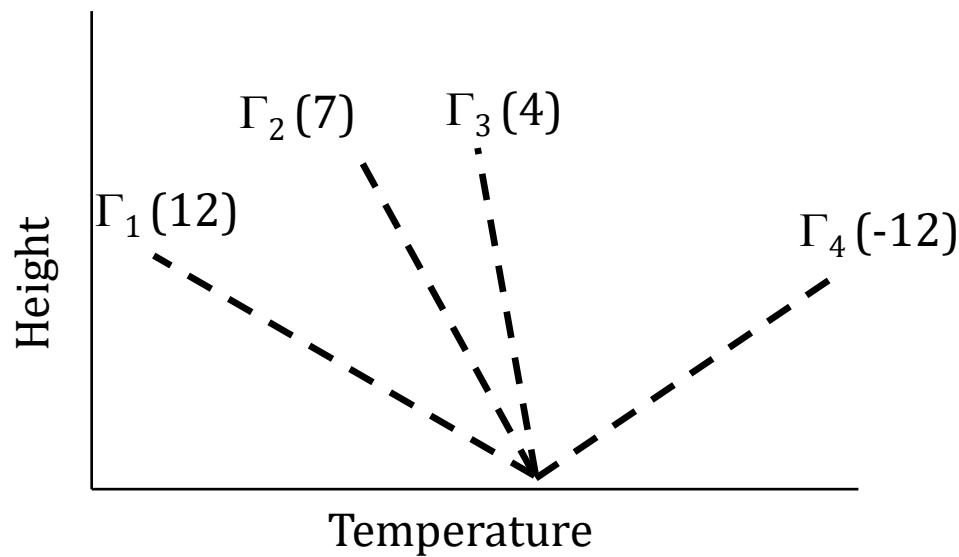
- The change in temperature with height

$$\Gamma = -\frac{\Delta T}{\Delta z}$$

Γ : lapse rate (K/km or °C/km).

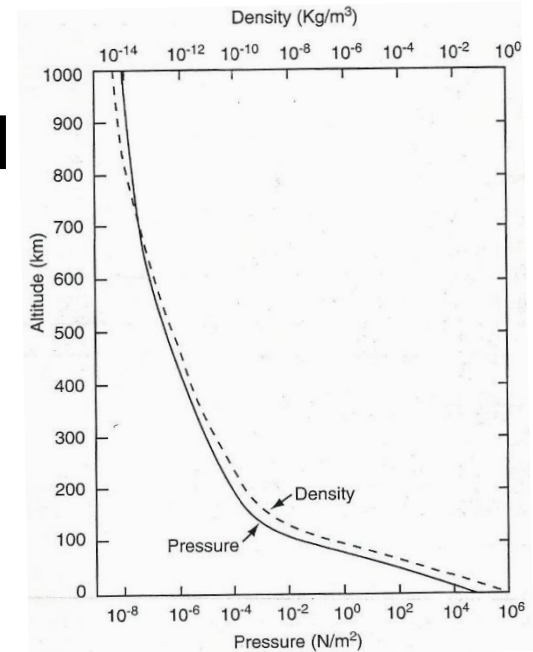
ΔT : temperature changes (K or °C) over height Δz (km)

negative if decreases with altitude increasing.



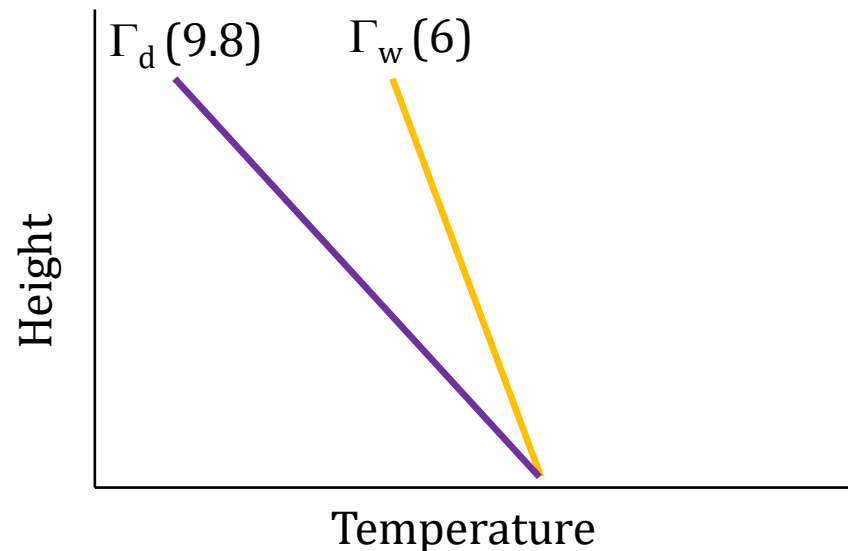
Rise of air parcel

- Suppose we have a parcel of air ascending from ground, *adiabatically*.....
 - Adiabatic Process: Process by which no energy is exchanged between a system (parcel of air) and its surroundings (atmosphere).
- Height $\uparrow$ pressure $\downarrow$ parcel expands
Air parcel $T \downarrow$
 - Due to energy balance
 - *No heat exchange between parcel & environment (adiabatic)*



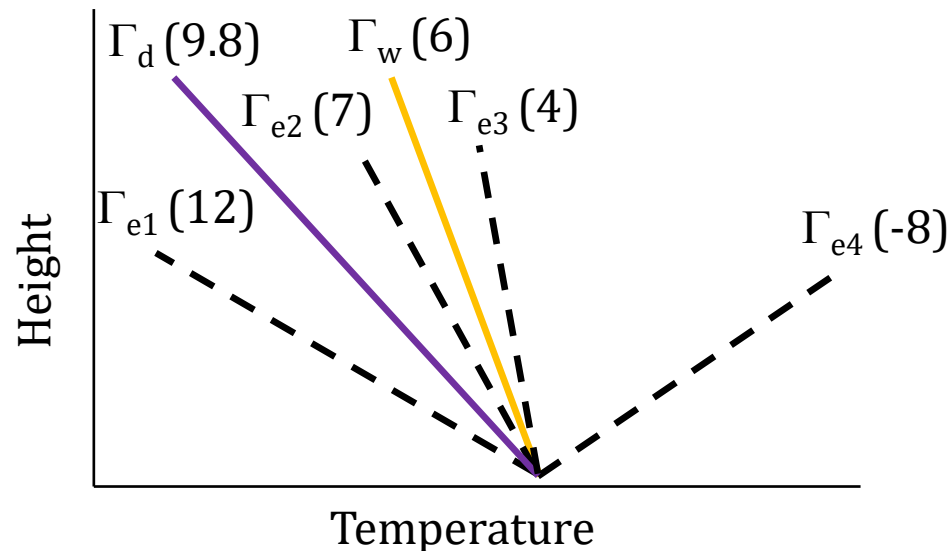
Rise of air parcel

- Dry air parcel: lose $9.8\text{ }^{\circ}\text{C}/\text{km} = \Gamma_d$
- Wet air parcel (saturated with moist) lose less temperature than dry air parcel
 - $T \downarrow$, moist $\rightarrow$ water $\rightarrow$ release heat
 - Γ_w varies $\approx 6\text{ }^{\circ}\text{C}/\text{km}$ in this class



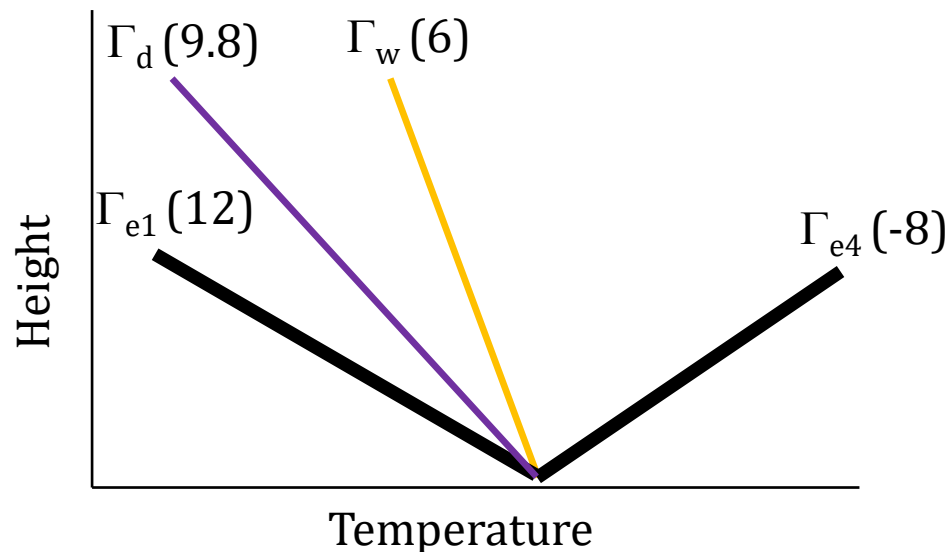
Rise of air parcel

- Lapse rate of lower troposphere
 - Environment lapse rate (Γ_e), varies greatly
- Air warmer than surrounding gain buoyancy
- How different Γ_e impact air parcel rising (vertical movement)?



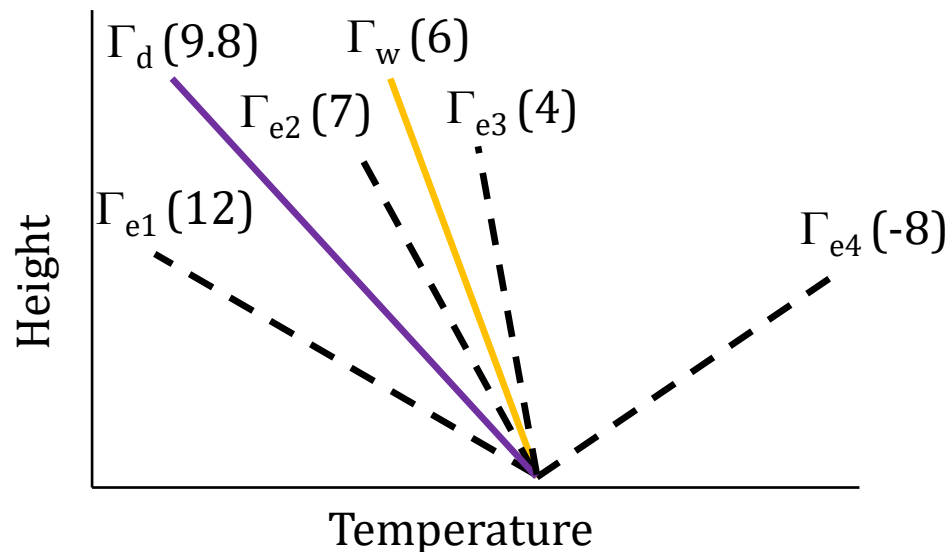
Rise of air parcel

- $\Gamma_e > \Gamma_d$: *Absolutely unstable* (Γ_{e1})
 - T of air parcel always > T of surrounding → keeps rising
- $\Gamma_e < \Gamma_w$: *Absolutely stable* (Γ_{e4})
 - T of air parcel always < T of surrounding → stop rising



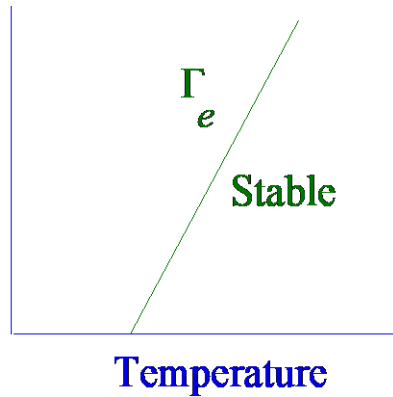
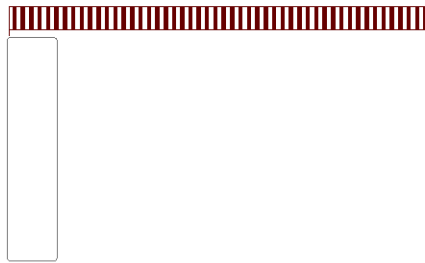
Rise of air parcel

- ☐ $\Gamma_e > \Gamma_d$: Absolutely unstable (Γ_{e1})
- ☐ $\Gamma_e = \Gamma_d$: Dry neutral
- ☐ $\Gamma_d > \Gamma_e > \Gamma_w$: Conditionally unstable (Γ_{e2})
- ☐ $\Gamma_e = \Gamma_w$: Wet neutral
- ☐ $\Gamma_e < \Gamma_w$: Absolutely stable (Γ_{e3}, Γ_{e4})

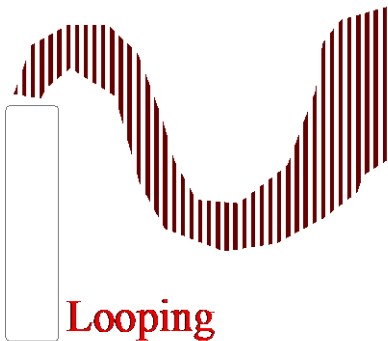
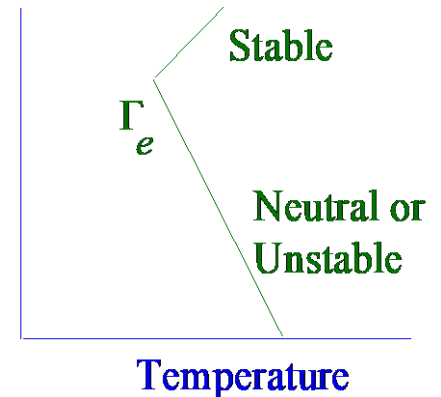
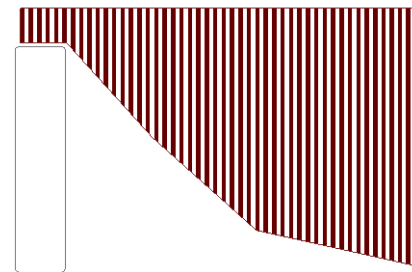


Effects on Pollutant Releases

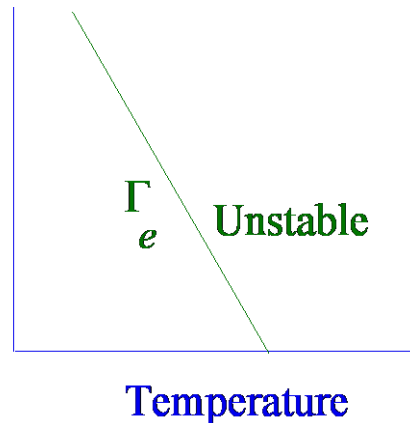
Fanning



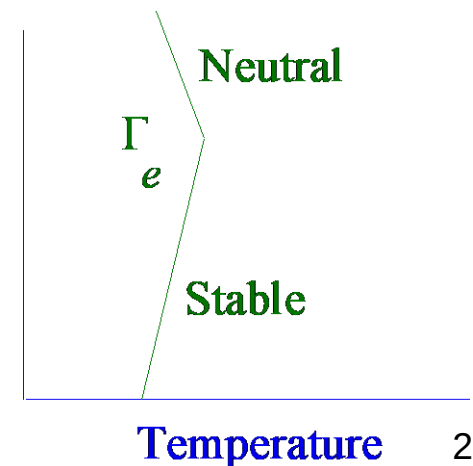
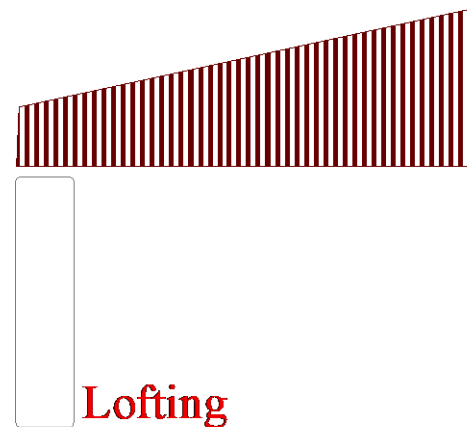
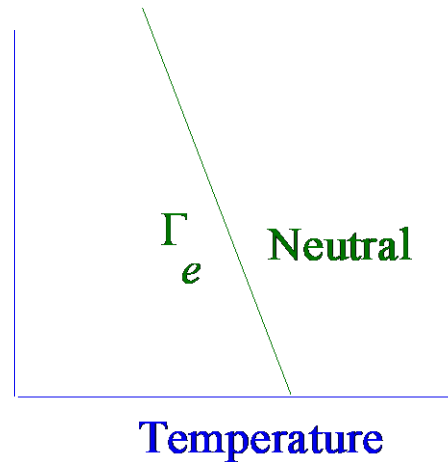
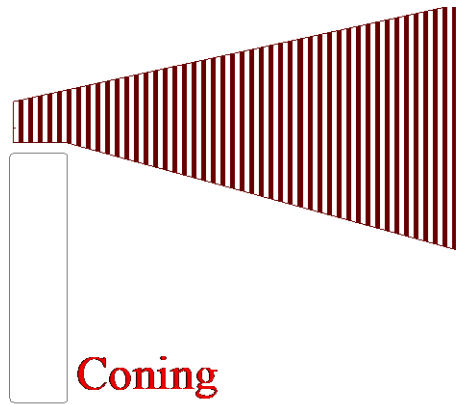
Fumigating



Looping

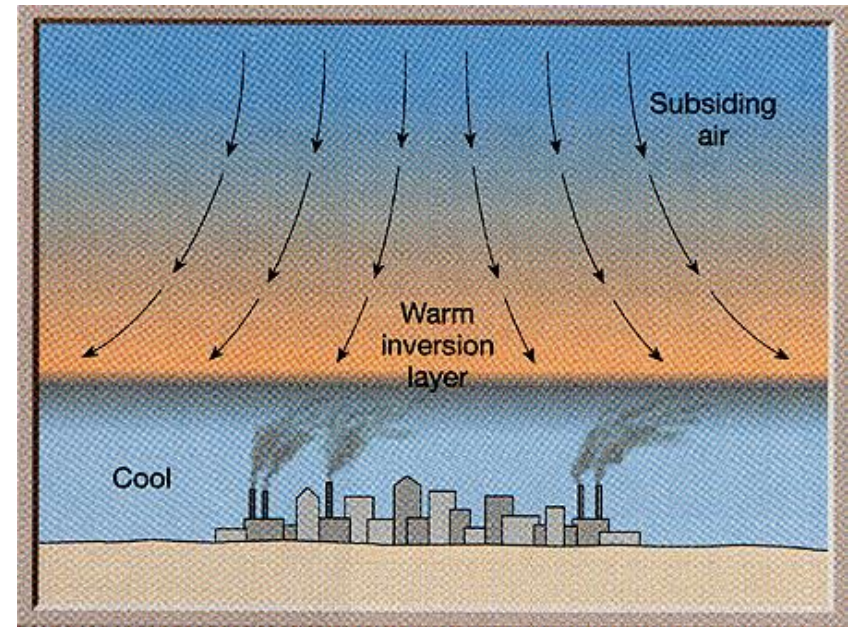


Effects on Pollutant Releases



Inversions

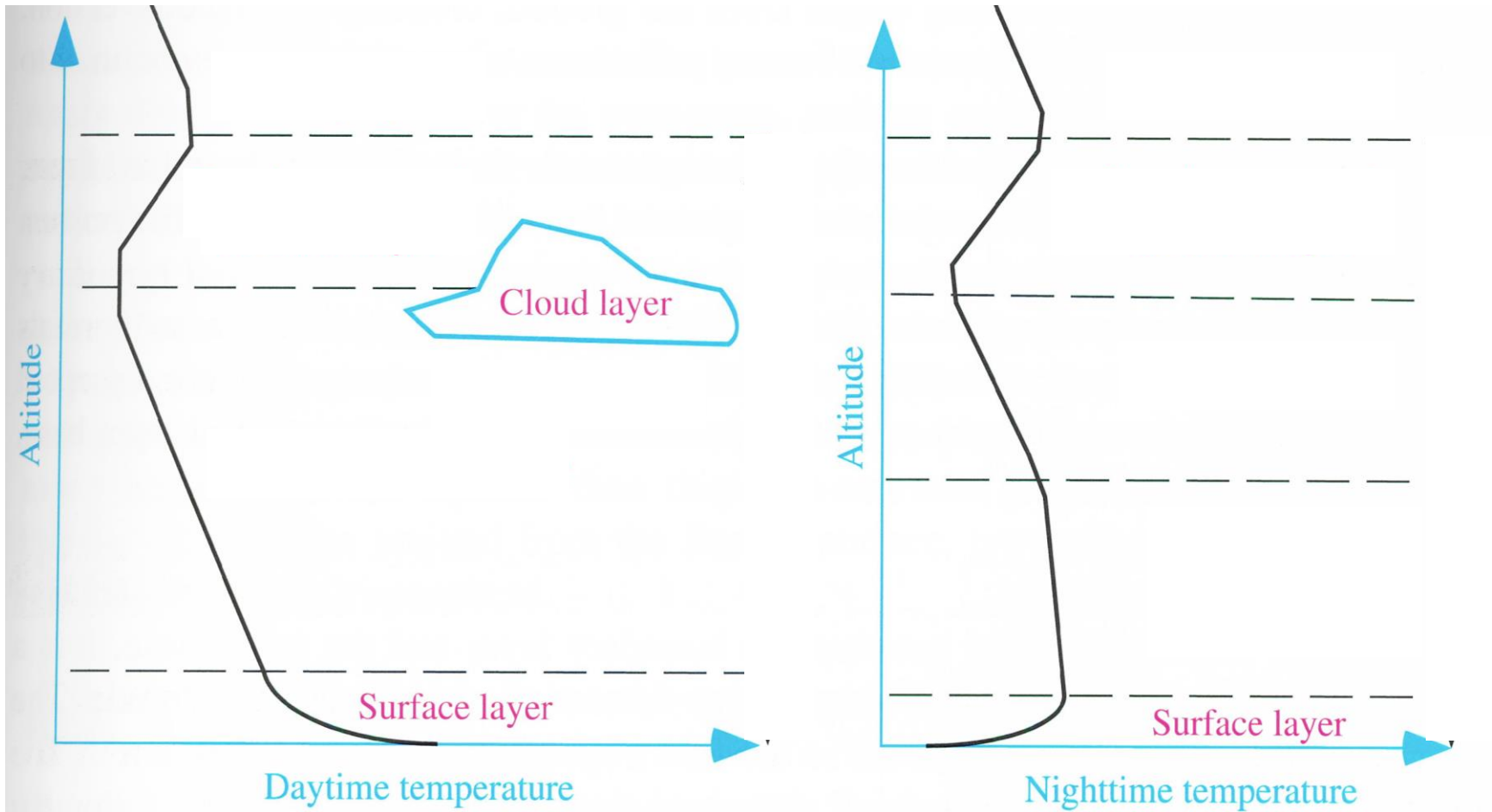
- Definition: a negative environmental lapse rate
 - Temperature increases with height
- Always stable
- Why important?
 - Can trap air and pollutants
- Types and effects
 - Frontal
 - Nocturnal (radiation)
 - Subsidence



http://apollo.lsc.vsc.edu/classes/met130/notes/chapter18/graphics/subsidence_inversion.gif

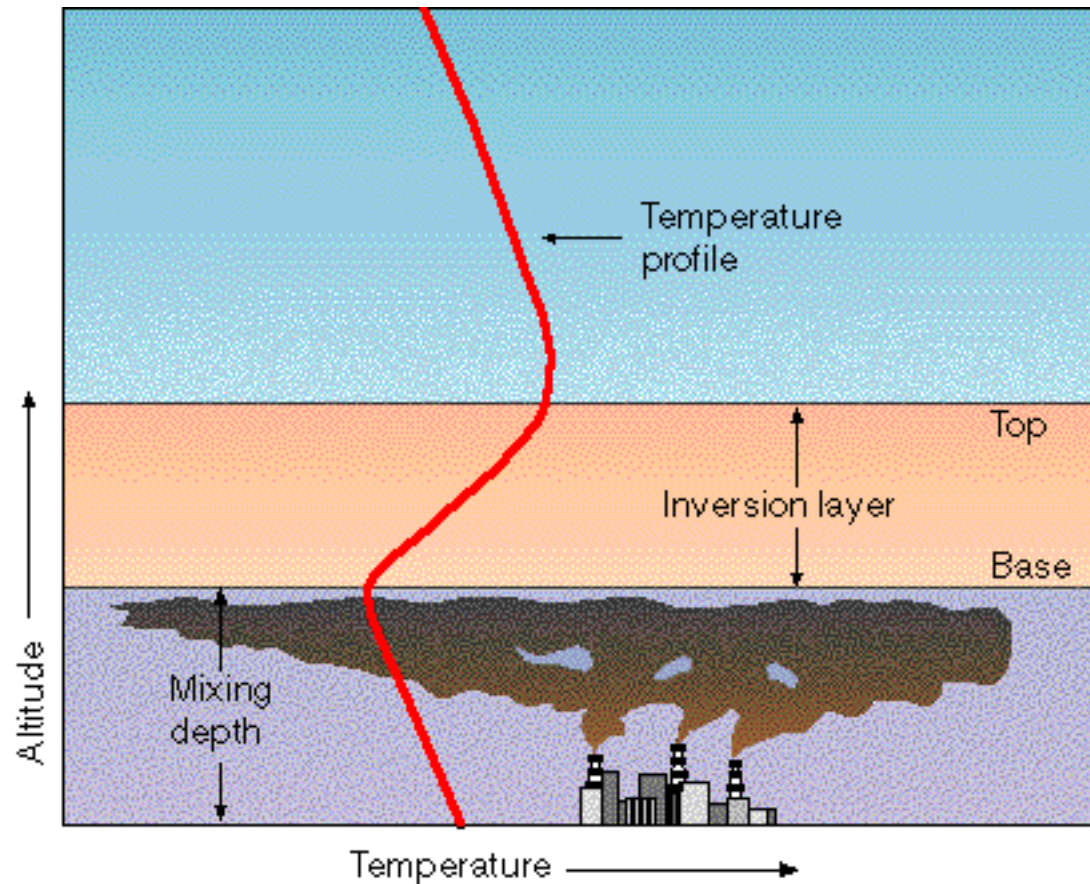
Tropospheric Temperature Structure

Diurnal Variation



Mixing Depth /Height

- Height to which air is vigorously mixed
- Definable if unstable or neutral below and stable above



Categorizing Stability: Classes

more stable with less clouds
→

day: more stable as u increases
night: less stable as u increases
↓

Surface Wind (measured at 10 m)	more stable with less sunlight → Day-Time Insolation			Night-Time Cloudiness	
m s ⁻¹	Strong	Moderate	Slight	Thinly Overcast or ≥ 4/8 Cloudiness	≤3/8 Cloudiness
<1	A	A-B	B	-	-
2-3	A-B	B	C	E	F
3-5	B	B-C	C	D	E
5-6	C	C-D	D	D	D
>6	C	D	D	D	D

A-extremely unstable, B-moderately unstable, C-slightly unstable,
D-neutral, E-slightly stable, F-moderately stable

(Hanna et al., 1982, Handbook on Atmospheric
Diffusion. U.S. Dept. of Energy).